Healing
شفاء

Doctor of Pharmacy 2016

Calculus 102A

Second exam

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#نبض_من_الشفاء_ارتوى

بسم الله الرحمن الرحيم

calculas

102A

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Second...

الحمد لله

30 علامة بالسكند

① Continuity الاتصال

Defn: A function f is said to be a continuous (متصلة) at $x=c$ if the following conditions are satisfied:

① $f(c)$ is defind. — ق(س) = عدد.

② $\lim_{x\to c} f(x)$ exist. — نهـا ق(س) موجودة.

③ $f(c) = \lim_{x\to c} f(x)$. — نهـا ق(س) = ق(س)

شروط الاتصال

EX: Determine if the following functions are continuous at $x=2$.

① $f(x) = \dfrac{x^2-4}{x-2}$

② $f(x) = \begin{cases} \dfrac{x^2-4}{x-2} & \text{if } x \neq 2 \\ 3 & \text{if } x = 2 \end{cases}$ (lim ← $x \neq 2$; $f(x)$ ← $x=2$)

③ $f(x) = \begin{cases} \dfrac{x^2-4}{x-2} & \text{if } x \neq 2 \\ 4 & \text{if } x = 2. \end{cases}$

Solution:

① $f(x) = \dfrac{(\cancel{x-2})(x+2)}{(\cancel{x-2})} = x+2$; $x \neq 2$

يجب كتابة الشرط فقط عندما تختصر الصورة

$\because f(x) = x+2$; $x \neq 2$

$f(2)$ is undefinded

so is $\underbrace{\text{dis continuous}}_{\text{غ متصل...}}$ at $x=2$

ملاحظة: في الصورة اذا اختصرنا صفر المقام تكون الصورة لصفر المقام يجب معرفة عددها.
غ مختزلة " " " " " " عدد عند صفر المقام.

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① $f(2)=3$

② $\lim_{x\to 2} f(x) =$

$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} \frac{x^2-4}{x-2} = 4$

$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} \frac{x^2-4}{x-2} = 4$

$= 4$

* لكن الطريقة الأفضل في حالة $(\neq)$، $(=)$.

$$\lim_{x\to 2} \frac{(x^2-4)}{(x-2)} = \lim_{x\to 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}} = 4$$

③ $f(2) \neq \lim_{x\to 2} f(x)$. so $f(x)$ is dis continous at $x=2$.

③ ⇒ نفس حل المثال الثاني لكن هذا المثال متصل عند $x=2$

H.w: $f(x) = \begin{cases} \frac{|x-2|}{x-2} & \text{if } x \neq 2. \\ 1 & \text{if } x = 2. \end{cases}$

1- $f(2)=1$

2- $\lim_{x\to 2} \frac{2-x}{x-2} = -1$

$\therefore \lim_{x\to 2^-} f(x) = -1$

$\lim_{x\to 2^+} \frac{x-2}{x-2} = 1$

$\therefore \lim_{x\to 2^+} f(x) = 1$

$\lim_{x\to 2} f(x) =$ isn't exist

so is dis continuous at $x=2$

EX: Find the value of a and b that.

$$f(x)=\begin{cases} 3x^2+ax & \text{if } x\le 1 \\ \dfrac{x^2-4}{x-2} & \text{if } 1< x<2 \\ 4b+2x & \text{if } x\ge 2 \end{cases}$$

is continous?

في كل نقطة تفرع نستطيع ايجاد الصورة والنهاية وبالتالي الحصول على معادلة.

※ من خلالها نستطيع ايجاد المجاهيل.

Sol: $\lim\limits_{x\to 1^+} f(x) = \lim\limits_{x\to 1^-} f(x)$

$$\lim_{x\to 1^+} \frac{x^2-4}{x-2} = \lim_{x\to 1^-} 3x^2+ax.$$

$$\frac{\cancel{-}3}{\cancel{-}1} = 3+a \Rightarrow \boxed{a=\text{zero}}$$

To find the value of b.

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x)$$

$$\lim_{x\to 2^-} \frac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}} = \lim_{x\to 2^+} 4b+2x$$

$$4 = 4b+4 \Rightarrow \boxed{b=\text{zero}}$$

② Derivative المشتقة

Defn: A function f is said to be differentiable at $x = a$ if:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \quad \text{exist}$$

Notation: $y = f(x)$

$\frac{dy}{dx} = f'(x)$.

Ex: Use definition of driveative to find $f'(x)$ for the following:

① $f(x) = x^2 + 4x$.

② $f(x) = \frac{1}{\sqrt{x}}$.

solution:

① $f'(x) = 2x + 4$

② $f(x) = (x)^{-1/2}$

$f'(x) = -\frac{1}{2}(x)^{-3/2} * 1$

Sol ②: $\acute{f}(x) = \lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h} \Rightarrow \lim\limits_{h \to 0} \dfrac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h}$

$$\lim_{x \to 0} \frac{\sqrt{x} - \sqrt{x+h}}{h(\sqrt{x})(\sqrt{x+h})} * \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}}$$

$$\Rightarrow \lim_{x \to 0} \frac{\cancel{x} - (\cancel{x} + h)}{\cancel{h}(\sqrt{x})(\sqrt{x+h})} * \frac{1}{\sqrt{x} + \sqrt{x+h}}$$

$$\Rightarrow \frac{-1}{(\sqrt{x})(\sqrt{x+h})(\sqrt{x} + \sqrt{x+h})}$$

$$\Rightarrow \frac{-1}{x(2\sqrt{x})} = \frac{-1}{2x^{3/2}} \Rightarrow \acute{f}(x) = \frac{-1}{2\sqrt{x^3}}$$

③ * Differentiation Rules قواعد الاشتقاق

	function	Derivative
①	$f(x) = K$ constant ثابت	$\acute{f}(x) = 0$
②	$f(x) = x^n$	$n\, f(x)^{(n-1)}$
③	$f(x) = g(x)\, h(x)$	$\acute{f}(x) = g(x)\, \acute{h}(x) + h(x)\, \acute{g}(x)$
④	$f(x) = \dfrac{g(x)}{h(x)}$	$\acute{f}(x) = \dfrac{h(x)\, \acute{g}(x) - [g(x)\, \acute{h}(x)]}{(h(x))^2}$ ← عشان الإشارة
⑤	$f(x) = \sqrt{g(x)}$	$\acute{f}(x) = \dfrac{\acute{g}(x)}{2\sqrt{g(x)}}$
⑥	$f(x) = \dfrac{a}{g(x)}$	$\acute{f}(x) = \dfrac{-a\, \acute{g}(x)}{(g(x))^2}$
⑦	$f(x) = g(x) \pm h(x)$	$\acute{f}(x) = \acute{g}(x) \pm \acute{h}(x)$

	function	Derivative
8	$f(x) = a\, g(x)$	$f'(x) = a\, g'(x)$
9	$f(x) = (g(x))^n$	$f'(x) = \underbrace{n\,(g(x))^{n-1}}_{\text{مشتقة القوس}} * \underbrace{(g'(x))}_{\text{مشتقة ما داخل القوس}}$
10	$y = f(u),\ u = f(x)$ $\frac{dy}{du}$ $\frac{du}{dx}$	$\frac{dy}{dx} = \frac{dy}{du} * \frac{du}{dx}$
11	$y = f(u),\ u = f(s),\ s = f(x)$ $\frac{dy}{du}$ $\frac{du}{ds}$ $\frac{ds}{dx}$	$\frac{dy}{dx} = \frac{dy}{du} * \frac{du}{ds} * \frac{ds}{dx}$

EX: Find $f'(x)$ or $\frac{dy}{dx}$ for.

① $y = x^5 + \frac{1}{\sqrt{x^3}} + 5$ تجميع

② $f(x) = \sqrt{x^5 + 4x}$

③ $f(x) = \frac{3}{(x^2 + 4x)}$

④ $f(x) = (x^5 + 4x^2 + 3x)^7$

⑤ $f(x) = \frac{x^2 + 4}{3x^5 + 4x}$

⑥ $f(x) = (\sqrt{x^2 + 4x})(3x^3 + 4)$

⑦ $f(x) = \frac{1}{\sqrt[3]{(x^4 + 2)^3 + 5x}}$

⑧ $y = u^3 + 4u$

$u = \sqrt{3x + 6}$

Sol $\Rightarrow$ $y = x^5 + (x)^{-3/2} + 5$

① $\Rightarrow$ $\frac{dy}{dx} = 5x^4 + 3/2 * (x)^{-5/2} + 0$

$\frac{dy}{dx} = 5x^4 - \frac{3/2 * 1}{\sqrt{x^5}}$

② $f'(x) = \frac{5x^4 + 4}{2\sqrt{x^5 + 4x}}$

③ $f'(x) = \frac{-3 * (2x + 4)}{(x^2 + 4x)^2}$

$f'(x) = \frac{-6x - 12}{(x^2 + 4x)^2}$

(4) $\Rightarrow f'(x) = 7(x^5+4x^2+3x)^6 * (5x^4+8x+3).$

عشان الإشارة

(5) $f'(x) = \frac{(3x^5+4x) * 2x - [(x^2+4)(15x^4+4)]}{(3x^5+4x)^2}$

$\vdots$

اذا طلب ... $f'(1) = -\frac{81}{49}.$

(6) $f'(x) = (\sqrt{x^2+4}) * (9x^2) + (3x^3+4) * \left(\frac{2x}{2\sqrt{x^2+4}}\right).$

(7) $f(x) = ((x^4+2)^3 + 5x)^{-1/3}$

$f'(x) = -\frac{1}{3} * ((x^4+2)^3 + 5x)^{-4/3} * (3(x^4+2)^2 * 4x^3 + 5).$

(8) By chain Rule $\Rightarrow$ باستخدام قاعدة السلسلة.

$\frac{dy}{dx} = \frac{dy}{du} * \frac{du}{dx} \Rightarrow (3u^2+4)\left[\frac{3}{2\sqrt{3x+6}}\right].$

$\frac{dy}{dx} = \left[3(\sqrt{3x+6})^2 + 4\right]\left[\frac{3}{2\sqrt{3x+6}}\right].$

$= [(9x+22)]\left[\frac{3}{2\sqrt{3x+6}}\right].$

EX: Let $f(x) = \sqrt{x^2+5}$ and $g(x) = \frac{1}{x^5+4x}$

Find $(f\circ g)'_{(x)}$, $(f\circ g)'_{(1)}$, $((f\circ g)_{(2)})'$

$\Downarrow$ zero.

Sol: $(f\circ g)_{(x)} = f(g(x))$

$(f\circ g)'_{(x)} = f'(g(x)) * g'(x).$

But:

$* \quad f'(x) = \frac{2x}{2\sqrt{x^2+5}} = \frac{x}{\sqrt{x^2+5}}.$

$g'(x) = \frac{-(5x^4+4).}{(x^5+4x)^2}$

But: $(f\circ g)'(x) = f'(g(x)) * g'(x)$

$= f'\left(\frac{1}{x^5+4x}\right) * \left[\frac{-(5x^4+4)}{(x^2+4x)^2}\right].$

$$= \frac{\frac{1}{x^5+4x}}{\sqrt{\left(\frac{1}{x^5+4x}\right)^2+5}} * \left[\frac{-(5x^4+4)}{(x^5+4x)^2.}\right]$$

$$(f\circ g)_{(1)} = \frac{\frac{1}{5}}{\sqrt{\frac{1}{25}+5}} * \frac{-9}{25}$$

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Healing 2016

4 * Equation of the tangent line.

معادلة المماس.

The Equation of the tangent line to $y = f(x)$ at (x_0, y_0).

. مهم

is : $y - y_0 = m(x - x_0)$.

where: slope $= m = \left.\frac{dy}{dx}\right|_{x = x_0} = f'(x_0)$.

Ex: Find the slope and the equation of the tangent line to $f(x) = \sqrt{x+5}$ at $x = 4$?

sol slope $m = f'(4) = \left.\frac{1}{2\sqrt{x+5}}\right|_{x=4} = \frac{1}{2\sqrt{4+5}} = \frac{1}{6}$

$x_0 = 4$

$y_0 = \sqrt{4+5} = 3$

Equation: $y - y_0 = m(x - x_0)$

$y - 3 = \frac{1}{6}(x - 4)$

$6y - 18 = x - 4$

$6y - x - 14 = 0$

* Higher derivatives   المشتقات العليا .. ⑤

If $y = f(x)$, then

- $\frac{dy}{dx} = f'(x)$
- $\frac{d^{2}y}{dx^{2}} = f''(x)$
- $\frac{d^{3}y}{dx^{3}} = f'''(x)$
- $\frac{d^{4}y}{dx^{4}} = f''''(x)$

EX: Find $\left.\frac{d^{3}y}{dx^{3}}\right|_{x=2}$ $f'''(2)$ for

H.W ① $y = \frac{x}{x+1}$

② $f(x) = \frac{1}{x}$

Sol(2): $f'(x) = \frac{-1}{x^{2}} \Rightarrow f''(x) = \frac{--1 \times (2x)}{x^{4}} = \frac{+2}{x^{3}}$

$$f'''(x) = \frac{-2 \times 3x^{2}}{x^{6}} = \frac{-6}{x^{4}}$$

$$\therefore f'''(2) = \frac{-6}{16} = \frac{-3}{8}.$$

⑥ * L'Hôpital Rule   قاعدة لوبيتال

* تذكر أن قاعدة لوبيتال لا تستخدم إلا في حالة كان ناتج $\frac{\text{صفر}}{\text{صفر}}$.

Then: Let $f(x)$ and $g(x)$ be continuous and differentiable (قابلة للاشتقاق ← تفاضل) function at $x=a$ and $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0}$ Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}.$$

EX: Use l'Hôpital Rule to find:

① $\lim_{x\to 1} \frac{x^{8}-1}{x^{2}-x} = \frac{0}{0}$

② $\lim_{h\to 0} \frac{(1+2h)^{20}-1}{h} = \frac{0}{0}$ ← نشتق بالنسبة لـ (h) لو كان في المسألة (x, y, ...) تعتبر ثابت.

Sol(1): $\lim\limits_{x \to 1} \dfrac{x^8-1}{x^2-1} \Rightarrow \lim\limits_{x \to 1} \dfrac{8x^7}{2x-1} = \dfrac{8}{2-1} = 8$

Sol(2): $\lim\limits_{h \to 0} \dfrac{20(1+2h)^{19} * 2 - 0}{1} = 20(1)^{19}(2) = 40$

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* Derivative of trigonometric function

الاقترانات الدائرية.

	function	Derivative ...
جا س	$f(x) = \sin x$	$f'(x) = \cos x$
جتا س	$f(x) = \cos x$	$f'(x) = -\sin x$
ظا س	$f(x) = \tan$	$f'(x) = \sec^2 x$
ظتا س	$f(x) = \cot x$	$f'(x) = -\csc^2 x$
قا س	$f(x) = \sec x$	$f'(x) = \sec x \tan x$
قتا س	$f(x) = \csc x$	$f'(x) = -\csc \cot x$.

* دائماً التي تبدأ بحرف (c) ← مشتقتها سالبة.

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* Derivat of Exponential (الأسية) and logrithmic (اللوغاريتم)...

مراس = لوس ب

$f(x) = \log_b^x$

مراس = لوس e = لطاس

$f(x) = \log_e^x = Lnx$

logrithm natural

function	Derivative
① $f(x) = e^x$	$f'(x) = e^x * (1)$ ← مشتقة الأس
② $f(x) = b^x$	$f'(x) = b^x * Ln\,b * (1)$
③ $f(x) = Lnx$	$f'(x) = \frac{1}{x}$ ← مشتق داخل اللوغاريتم / داخل اللوغاريتم
④ $f(x) = \log_b x$	$f'(x) = \left(\frac{1}{x}\right) * \frac{1}{Ln\,b}$
⑤ $f(x) = Ln\,g(x)$	$f'(x) = \frac{g'(x)}{g(x)}$ ← مشتق داخل / داخل

Ex: Find $f'(x)$ for:

① $f(x) = (\cos x)(x^2+4x)$.
② $f(x) = \sec^3 x \Rightarrow (\sec\ )^3$
③ $f(x) = \tan(\sin x)$.
④ $f(x) = 3^x$.
⑤ $f(x) = e^{\sin x}$.
⑥ $f(x) = \ln(x^2+4x)$.
⑦ $f(x) = \log_3 \sin x$.
⑧ $f(x) = \ln(\cot x)$.
⑨ $f(x) = (\csc x)(x^2+4)$.

Solution:

① $f'(x) = (\cos x)(2x+4) + (x^2+4x) * (-\sin x)$.

② $f'(x) = 3\sec^2 * (\sec x \tan x) = 3\sec^3 \tan x$.

③ $f'(x) = \sec^2(\sin x) * (\cos x)$

④ $f'(x) = 3^x \ln 3$

⑤ $f'(x) = e^{\sin x} * [\cos x]$.

⑥ $f'(x) = \frac{2x+4}{x^2+4x}$.

⑦ $f'(x) = \left(\frac{\cos x}{\sin x}\right) * \frac{1}{\ln 3}$.

⑧ $f'(x) = \frac{-\csc^2 x}{\cot x}$.

⑨ $f'(x) = (\csc x)(2x) + (x^2+4)(-\csc x \cot x)$.

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* previous Exams (Multiple choice)

① Let $f(x) = x^4 - x - 1$ find $(f \circ f)'(1)$?

A) -5 B) -3 C) 5 D) -15

Sol: step (1): $(f \circ f)(x) = f(f(x))$

step (2): $f'(f(x)) \times f'(x)$

step (3): $f'(f(1)) \times f'(1)$

But: $f(1) = -1$

$f'(x) = 4x - 1$

$f'(1) = 4 - 1 = 3$

$f'(-1) = 5$

$f'(-1) \times 3 \Rightarrow -5 \times 3 = -15$

② let $f(x) = (3x-2)^{20}$. Then $\lim_{h \to 0} \frac{f(1+h) - f(1)}{h^2 + 2h} = \frac{0}{0}$

A) 10 B) 20 C) 30 D) 40 E) 60

By L'Hôpital Rule

solution: * $\lim_{h \to 0} \frac{f'(1+h) \times 1 - 0}{2h + h} = \frac{f'(1)}{2}$

But: $f'(x) = 20(3x-2)^{19} \times 3 \Rightarrow f'(1) = 60$

$\therefore \lim_{h \to 0} \frac{f'(1)}{2} = \frac{60}{2} = 30$

* $\lim_{h \to 0} \frac{f(1+h) - f(1)}{h(h+2)} = \lim_{h \to 0} \frac{f'(1)}{(h+2)}$

$= \frac{f'(1)}{2} = \frac{60}{2} = 30.$

③ Let $f(x) = \underline{(g(x)+3)}\,\underline{\ln(x+1)}$. $g(0)=2$

Find $f'(0)$.

Solution: $f'(x) = (g(x)+3) * \left(\frac{1}{x+1}\right) + \ln(x+1)\,(g'(x)+0)$.

$f'(0) = (g(0)+3) \times \frac{1}{0+1} + \ln(1) * (g'(0)+0)$

$5 + 0 \times (---)$

$= 5$

H.W

④ let $y = 2\cos(t)$, $t = \frac{\pi}{2} * x^2$. find $\frac{dy}{dx}$ $x=1$

$\frac{dy}{dx} = \frac{dy}{dt} * \frac{dt}{dx}$

$\frac{dy}{dx} = \frac{dy}{dt} * \frac{dt}{dx}$

$= (-2\sin(x))(\pi x)$.

$= -2\pi x \sin(t)$.

But, whe $x=1$:

$= -2\pi * \sin\left(\frac{\pi}{2}\right)$

$= -2\pi * 1 = -2\pi$

But:

$* \frac{dt}{dx} = \frac{2\pi}{2}x = \boxed{\pi x}$

$* \frac{dy}{dt} = -2\sin(t)$.

$t = \frac{\pi}{2}$

⑤ let $f(x)=$

⑦ Let $f(x) = \ln x + e^{\sin(\pi x)}$ find $f'(1)$.

A) $1-\pi$ B) $1+\pi$ C) 0 D) π.

Solu: $f'(x) = \frac{1}{x} + e^{\sin(\pi x)} * \cos(\pi x) * \pi$

$f'(1) = 1 + e^{0} * \cos(\pi) * \pi$

$= \cancel{1+e}\ 1 + 1 * -1 * \pi$

$* \ = 1 - \pi$

⑧ Let $y = \frac{x-1}{x+1}$. find $\frac{d^3y}{dx^3}$.

سوف نخطئها، متقبلة خاف لأننا أخطأنا

طبعاً وأنا كمان اتخبطت

$y = (x-1)(x+1)^{-1}$

$\frac{dy}{dx} = (x-1)(-1*(x+1)^{-2}) + (x-1)^{-1} * 1 \quad = (1-x)(x+1)^{-2} + (x+1)^{-1}$

$\frac{d^2y}{dx^2} = (x-1)(\overset{+}{\cancel{-}}2*(x+1)^{-3}) + (-1(x+1)^{-2}) * (1) \quad + (-1*(x-1)^{-2}$

$= 2(x-1)(x+1)^{-3} - (x+1)^{-2} - (x-1)^{-2}$

$= 2(x-1)(x+1)^{-3} - (x+1)^{-2} - (x-1)^{-2}$

$\frac{d^3y}{dx^3} = (2(x-1)*-3(x+1)^{-4} + (x+1)^{-3} \times 2\cancel{*}) - (-2(x+1)^{-3}) - (-2(x-1)^{-3}$

$\cancel{=2(x-1)}$

$= -6(x-1)(x+1)^{-4} + 2\cancel{*}(x+1)^{-3} + 2(x+1)^{-3} + 2(x+1)^{-3}$

$-6(x-1)(x+1)^{-4} + 6(x-1)^{-3}$

$= \cancel{6(x-1)(x+1)^{-4}} \Rightarrow 6(x+1)^{-3}(-(x-1)(x+1)^{-1} + 1)$

$= 6(x+1)^{-3}\left(\frac{-(x-1)}{x+1} + \frac{(x+1)}{(x+1)}\right) \Rightarrow 6(x+1)^{-3}\left(\frac{-x+1+x+1}{x+1}\right)$

$= 6(x+1)^{-3}\left(\frac{2}{x+1}\right) \quad = 12(x+1)^{-4}$

OR : باستخدام القسمة الطويلة

$y = \frac{x-1}{x+1} = 1 + \frac{-2}{x+1}$

$$\begin{array}{r|l} & 1 \\ (x+1) & (x-1) \\ & -x+1 \\ \hline & -2 \end{array}$$

⋮

$y'' = \frac{-4}{(x+1)^3} \Rightarrow y''' = \frac{12}{(x+1)^4}$

9. The derivative of $f(x) = \frac{1}{(\sqrt{x}+1)^2}$ is.

$\hookrightarrow f'(x) = \frac{-1 \times 2(\sqrt{x}+1) \times \frac{1}{2\sqrt{x}}}{(\sqrt{x}+1)^4} \Rightarrow \frac{-1}{\sqrt{x}(\sqrt{x}+1)^3}$

10: If $u = \sqrt{t}$, $t = \ln x + x$, then $\frac{du}{dx}$ at $x=1$

$\frac{du}{dx} = \frac{du}{dt} \times \frac{dt}{dx} \Rightarrow \frac{du}{dx} = \left(\frac{1}{2\sqrt{t}}\right) * \left(\frac{1}{x} + 1\right)$

But: $t = \ln(1) + 1 = 1$

$\frac{du}{dx} = \frac{1}{2} \times 2 = 1$

11. Let $y = 3^{\sin x} + \sec^2 x$ find $f'(0)$.

$y' = 3^{\sin(x)} * \ln 3 * (\cos x) + 2\sec * \sec * \tan(x).$

$\ln 3$

12. Find the slope of $y = 2^{\sin x}$ at $x = \pi$

slope $= y' = 2^{\sin(x)} * \ln 2 * \cos(x)$

$\ln 2$.

EX: Find the equation of the tangent line to:

① $f(x) = \frac{2x+1}{x+1} + \sin^3(x)$ at $x=0$.

② $f(x) = \sin^3(x)$ at $x = \frac{\pi}{4}$.

سؤال كتابة
وليس ضع
دائرة.

①

*

② Equation : $(y - y_0) = m(x - x_0)$

$x_1 = \frac{\pi}{4}$, y_1 , $m = f'(x_1)$.

But:

$y_1 = f(\frac{\pi}{4}) = \sin^3(\frac{\pi}{4}) = (\sin(\frac{\pi}{4}))^3$

$= (\frac{1}{\sqrt{2}})^3 = \frac{1}{2\sqrt{2}}$

$f'(x) = 3\sin(x)^2 * \cos(x)$.

$f'(\frac{\pi}{4}) = 3\sin(\frac{\pi}{4})^2 * \cos(x)$

$= 3(\frac{1}{\sqrt{2}})^2 * (\frac{1}{\sqrt{2}})$

$= 3 \times \frac{1}{2} \times \frac{1}{\sqrt{2}} = \frac{3}{2\sqrt{2}}$

$$\boxed{\therefore \ (y - \frac{1}{2\sqrt{2}}) = \frac{3}{2\sqrt{2}}(x - \frac{\pi}{4}).}$$

Question (5 point) :- Find $f'(x)$ for $f(x) = \ln\left(\frac{\cos x \sqrt{x^2+1}}{x^3}\right)$ using logarithm Rules.

Recall * $\ln(x * y) = \ln x + \ln y$.

* $\ln(\frac{x}{y}) = \ln x - \ln y$.

* $\ln(x^n) = n \ln x$.

 → → →

<u>Solution</u>: $f(x) = \ln(\cos x) + \ln\sqrt{x^2+1} - \ln x^3$

$$= \ln(\cos x) + \tfrac{1}{2}\ln(x^2+1) - 3\ln x$$

$$= \left(\frac{-\sin x}{\cos x}\right) + \frac{1}{2} * \left(\frac{2x}{x^2+1}\right) - \frac{3*1}{x}$$

$$= -\tan(x) + \frac{x}{x^2+1} - \frac{3}{x}.$$

صفحة $\boxed{١٩}$

Mohammad Al-Zoubi

Healing 2016

Domain of Exp. log and Trig. function:-

Remark:

* ① Domain of $f(x) = \left\{\begin{matrix} \sin g(x) \\ \cos g(x) \end{matrix}\right\}$ = Domain of $g(x)$

الزاوية

* ② Domain of $f(x) = b^{\boxed{g(x)}}$ = Domain of $g(x)$.

③ Domain of $f(x) = \ln(g(x)) = \{x : g(x) > 0\}$

ماداخل اللوغاريتم يجب ان يكون (+).

EX: find the following domain.

① $f(x) = \sin\left(\frac{1}{x}\right) \rightarrow$ Domain: $\mathbb{R} - \{0\}$.

② $f(x)\ 2^{\frac{x+1}{x-1}} \rightarrow$ Domain: $\mathbb{R} - \{1\}$.

أنشطة منوعة

③ $f(x) = \ln\left(\frac{1-x}{x}\right)$

④ $f(x) = \ln(3-x) - \ln(x+1)$.

⑤ $f(x) = \frac{\sqrt{x}}{2^x - 4}$

③ Domain of $f(x) = \left\{x : \frac{1-x}{x} > 0\right\}$.

$\frac{1-x}{x} > 0$

sign of $\frac{1-x}{x}$: $-$ at 0 (u), $+$ between 0 and 1, $-$ after 1 (0)

$(0, 1)$

(5)

Domain of $f(x)$ = Domain $\{\sqrt{x}\}$ $\cap$ Domain $\{2^x - 4\}$ $-$ $\{x : 2^x - 4 = 0\}$.

$x \geq 0 \quad \cap \quad \mathbb{R} - \{2\}$.

$\underbrace{[0, \infty) \cap \mathbb{R}}$ $- \{2\}$.

$[0, \infty) - \{2\}$ OR $[0, 2) \cup (2, \infty)$.

(4)

Domain of $f(x)$ = Domain $\{\ln(3-x)\}$ $\cap$ Domain $\{\ln(x+1)\}$.

OR

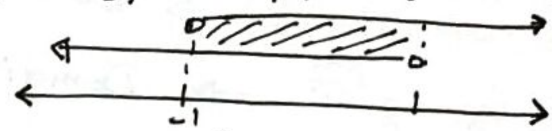
$\ln\left(\frac{3-x}{x+1}\right)$

$\frac{3-x}{x+1} > 0$

Domain is : $(-1, 3)$.

$3 - x > 0$ $\cap$

$(-\infty, 3) \quad \cap \quad (-1, \infty)$



Domain of $f(x)$ is $(-1, 3)$.

Solving Equations involving Exp. and log Expressions

Exponential Rule and properties:-

1. $\boxed{b^m * b^n} = \boxed{b}^{m+n}$
2. $\frac{\boxed{b^m}}{\boxed{b^n}} = b^{m-n}$
3. * $(b^m)^n = b^{mn}$
4. $b^{-m} = \frac{1}{b^m}$
5. $\left(\frac{b}{a}\right)^{-m} = \left(\frac{a}{b}\right)^m$
6. $(ab)^m = a^m b^m$
7. $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$
8. الجذر الأس
   $b^{m/n} = \sqrt[n]{b^m}$
9. $b^0 = 1$
   $b \neq 0$.

* Logarithm Rules and properties.

① $\log_b(xy) = \log_b(x) + \log_b(y)$.

② $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$.

③ $\log_b x^{n} = n\log_b x$.

*** ④ $b^{\log_b x} = x$. → Ex: $3^{\log_3 x} = x$

⑤ $\log_b b^x = x$. → Ex: $\log_3 3^x = x$.

⑥ $\log_b b = 1$, $\log_b 1 = 0$, $\ln e = 1$, $\ln 1 = 0$

*** ⑦ $\log_b a = \dfrac{\log_{\boxed{c}} a}{\log_{\boxed{c}} b} = \dfrac{\ln_e a}{\ln_e b}$

بقدر أفصلها لـ (log) أي أساس.

أي متين الماهم...

⑧ $\log_c a * \log_b c = \log_b a$

*** *** ⑨ $\log_{b^{n}} b^{m} = \dfrac{m}{n}$

⑩ <u>Golden Rule</u> : $y = \log_b x \Leftrightarrow b^y = x$

قوة

الأساس

ما داخل اللوغاريتم

Exp 9: $\Rightarrow \log_9 \sqrt{27} = \log_{3^2} 3^{3/2} = \dfrac{3/2}{2} = \dfrac{3}{4}$.

Exp 10: ~~to~~ $8 = \log_2 x \Rightarrow x = 2^8 = 256$

************** **EX(1): Find the Exact value:-**

أسئلة سنوات :

1. $\log_{2} 512$
2. $\log_{27} 243$
3. $\log_{10} 100$
4. $\log_{2} 0.125$
5. $\dfrac{\log_{2} 20 - \log_{2} 5}{\log_{2} 3 - \log_{2} 6}$
6. $(\sqrt{3})^{\log_3 9}$
7. $\log_{1/3} 27$
8. $3^{\log_{1/9} 16}$
9. $(\sqrt{2})^{\log_2 81}$
10. $\log_2 3 + \log_2 6 - \log_2 (9/\sqrt{2})$

Solution :~

اذا كانت اول منزلة زوجية فهو يقبل القسمة على 2 .

2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

1. $\log_{2^1} 2^9 = \frac{9}{1} = 9$

- * اوله زوجي او صفر ⇐ يقبل القسمة على (2).
- * اذا كان مجموع ارقامه يقبل القسمة على (3) . فانه يقبل على (3).
- * اذا كان اوله 0 او 5 فانه يقبل القسمة على 5 .

2. $\log_{3^3} 3^5 = \frac{5}{3}$

3. $\log_{10^1} 10^2 = 2$

4. $\log_2 \frac{125}{1000} = \log_2 1/8 = \log_2 2^{-3} = -3$

5. $\log_{3/6} \frac{20}{5} = \log_{1/2} 4 = \log_{2^{-1}} 2^2 = -2$

6. $\rightarrow (\sqrt{3})^{\log_3 3^2} = (\sqrt{3})^{2\log_3 3} = (\sqrt{3})^2 = 3$

OR

اولاً

$\rightarrow (\sqrt{3})^{\log_3 9} = 3^{(1/2)\log_3 9} = 3^{\log_3 9^{(1/2)}} = \sqrt{9} = 3$

⑧ $3^{\log_{1/9} 16} = 3^{2\log_{1/9} 4} = \cancel{9^{\log_9 4}}$

$= \left(\frac{1}{9}\right)^{-1 \log_{1/9} 4} = 4^{-1} = \frac{1}{4}$

$\cancel{3^{\log_3 2^{16}}}$

⑨ $(\sqrt{2})^{\log_2 81} = (\sqrt{2})^{\log_2 2^9} = (\sqrt{2})^9 = \text{-----}$

$2^{(1/2)\log_2 81} = 2^{\log_2 \sqrt{81}} = \sqrt{81} = 9$

Ex(2): Solve for x.

① $\log_2 x + \log_2 (x+2) = 3$.

② $\log_2 (\log_3 (x+1)) = 2$.

*③ $e^{2x} = 9$.

④ $2^{3x+4} = \frac{1}{16^x}$.

⑤ $3^x = 4$

⑥ $\log_3 (x+1) = 2$

⑦ $\ln x = 3$

$y = \log_b x \Leftrightarrow x = b^y$

Golden Rule.

---

① $\log_2 x(x+2) = 3 \Rightarrow 2^3 = x^2 + 2x$

$x^2 + 2x - 8 = 0 \Rightarrow (x+4)(x-2) = 0$

$x = -4$, $\boxed{x = 2}$

بعد التعويض بالمعادلة الأصلية .

② $\log_3 (x+1) = 2^2 \Rightarrow \log_3 (x+1) = 4$

$\Rightarrow x+1 = 3^4 = 81$

$x+1 = 81 \Rightarrow \boxed{x = 80}$

③ $(e^x)^2 = 9 \Rightarrow (e^x)^2 - 9 = 0 \Rightarrow (e^x - 3)(e^x + 3) = 0$

$e^x = 3$ $\qquad$ $e^x = -3$ ✗

④ $2^{3x+4} = \frac{1}{(2^4)^x} = 2^{-4x}$

$\Rightarrow 3x + 4 = -4x$

$7x = -4 \rightarrow x = \frac{-4}{7}$

⑤ $x = \log_3 4$

⑥ $x+1 = 3^2 \Rightarrow$ $\Rightarrow x+1 = 9$

$* \boxed{x = 8}$

⑦ $\ln x = 3 \Rightarrow \log_e x = 3 \Rightarrow x = e^3$

---

⑧

موضوع جديد

* limits of trig... function.

نهاية الاقترانات الدائرية.

Thm: $\lim_{x \to 0} \frac{\sin x}{x} = 1$

Remark

① $\lim_{x \to 0} \frac{\sin ax}{x} = a$

② $\lim_{x \to 0} \frac{\sin ax}{\sin bx} = \frac{a}{b}$

③ $\lim_{x \to} \frac{\tan ax}{x} = a$

④ $\lim_{x \to 0} \frac{\sin ax}{\tan bx} = \frac{a}{b}$

⑤ $\lim_{x \to 0} \frac{\tan ax}{\sin bx} = \frac{a}{b}$

⑥ $\lim_{x \to 0} \frac{\tan ax}{\tan bx} = \frac{a}{b}$

(1) **EX:** Find:

① $\lim_{x \to 0} \frac{\sin 3x}{x} = 3$

② $\lim_{x \to 0} \frac{\sin 4x}{\tan 5x} = \frac{4}{5}.$

منوات. ③ $\lim_{\theta \to 0} \frac{\tan^2 4\theta + \sin 5\theta}{\theta^2} \Rightarrow \text{limit} \frac{\frac{\tan^2 4\theta}{\theta^2} + \frac{\sin 5\theta}{\theta^2}}{\frac{\theta^2}{\theta^2}}$

④ $\lim_{x \to 0} \frac{\sin^2 4x}{\boxed{x}} =$ ~~$\lim 4x \cdot \sin 4x$~~ $\lim_{x \to 0} \frac{\sin 4x}{x} \cdot \sin 4x = 4 \times 0 = 0$

⑤ $\lim_{x \to \pi} \frac{\sin \boxed{2} x}{x} = \frac{\sin 2\pi}{\pi} = \frac{0}{\pi} = 0$

انتبه..

Trig Identites متطابقات

① $\sin^2 x + \cos^2 x = 1$

$2\frac{\sin x^{2}_{0}}{0}$

قسمنا $\cos^2$ ② $\tan^2 x + 1 = \sec^2$

قسمنا $\sin^2$ ③ $1 + \cot^2 x = \csc^2$

④ $\sin(2x) = 2\sin(x)\cos(x)$.

⑤ $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$. $\rightarrow 2\sin^2 x = 1 - \cos 2x$

⑥ $\cos^2 = \frac{1}{2}(1 + \cos 2x)$. $\rightarrow 2\cos^2 = 1 + \cos 2x$.

·(2) EX: Find:-

① $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$

② $\lim_{x \to 0} \frac{1 - \cos 5x}{1 - \cos 7x}$.

سنوات. ③ $\lim_{\theta \to 0} \frac{1 - \cos 2\theta}{\theta^2}$

④ $\lim_{x \to 0} \frac{1 + 2\sin x + \sin^2 x}{\sqrt{4 + \cos^2 x}}$.

بالتعويض المباشر

$= \frac{1}{\sqrt{5}}$

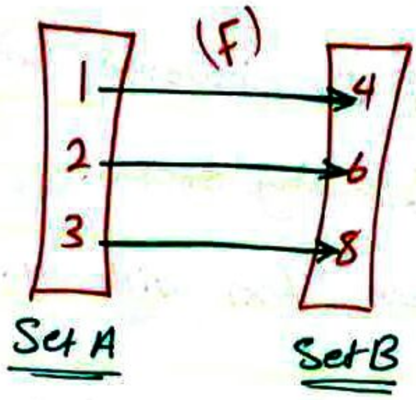
① $\lim_{x \to 0} \frac{1 - \cos x}{x^?} * \frac{1 + \cos x}{1 + \cos x} = \lim_{x \to 0} \frac{\sin x^2}{x^2(1 + \cos x)}$

$= \frac{1}{2}$

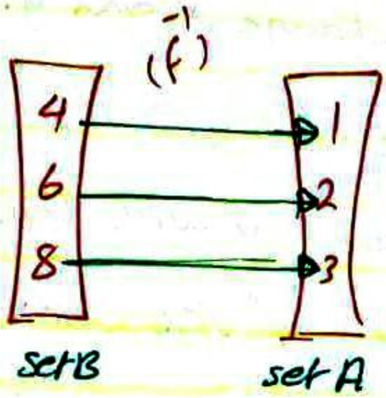
② $\lim_{x \to 0} \frac{2\sin^2 \frac{5}{2}x}{2\sin^2 \frac{7}{2}x} = \frac{\frac{25}{4}}{\frac{49}{4}} = \frac{25}{49}$.

③ $\lim_{\theta \to 0} \frac{1 - \cos 2\theta}{\theta^2} = \lim_{\theta \to 0} \frac{2\sin^2(\theta)}{\theta^2} = 2$

الاقتران العكسي. Inverse function $f^{-1}(x)$



* Domain of $f = A$.
* Domain of $f = B$.
* $f(1) = 4$, $f(2) = 6$, $f(3) = 8$
* $f(x)$ is one-to-one.
  ↳ $f(1) =$ س



* Domain of $f^{-1} = B$.
* Domain of $f^{-1} = A$.
* $f^{-1}(4) = 1$, $f^{-1}(6) = 2$ ...
* $f^{-1}(x)$ is one-to-one.

* properties of Invers function:-

① f and f^{-1} are one-to-one. function.
② Domain of $f^{-1}(x) =$ Range $f(x)$.
③ Range of $f^{-1}(x) =$ Domain of $f(x)$.
④ $f(f^{-1}(x)) = x$; $x \in$ Domain f^{-1}.
⑤ $f^{-1}(f(x)) = x$; $x \in$ Domain $f(x)$.

Caution $\left\{ f^{-1}(x) \neq \frac{1}{f(x)} \right\}$

? How can we find the inverse function.

Answer: we have to following steps:

① **step(1):** Replace $f(x)$ with y.

step(2): Interchange x and y.

خطوات حل هذه الأسئلة:

step(3): solve for y.

step(4): Replace y with $f^{-1}(x)$.

Ex: Find $f^{-1}(x)$ and Range of $f(x)$ for:

① $f(x)=\frac{9x-1}{3x-6}$ ← على طريقة القديمة

Domain $= R-\{2\}$.

Range $= R-\{3\}$.

لكن هنا المجال أصبح مدى والعكس المدى أصبح مجال.

② $f(x)=\log_2(x+4)$.

③ $f(x)=2^x-3$.

H.W ④ $f(x)=\frac{e^{2x}+3}{3e^{2x}-1}$

solution ② → step(1): $y=\log_2(x+4)$.

→ step(2): $x=\log_2(y+4)$.

→ step(3): $y+4=2^x \Rightarrow y=\underbrace{2^x}_{R}-\underbrace{4}_{R}$.

→ (4): Domain of f = Domain of f^{-1} = R

$- \quad \underset{-3}{|} \quad +$

$[-3, \infty)$

③ $y = 2^x - 3$

$x = 2^y - 3$

$2^y = x + 3$

$\log_2 2^y = \log_2 (x+3) \Rightarrow y = \log_2 (x+3)$

ما داخل اللوغاريتم يجب انه يكون (+). $\rightarrow$ $x+3 > 0$

$- - - - \underset{-3}{|} + + + + +$

Domain $= [-3 , \infty)$.

④ $\rightarrow y = \dfrac{e^{2x} + 3}{2e^{2x} - 1}$.

$\rightarrow x = \dfrac{e^{2y} + 3}{2e^{2y} - 1}$.

$\rightarrow$

لاحظ لانستطيع فصل x عن y.

Ex: let $f(x) = x^3 + x + 1$ find.

① $f^{-1}(11)$ ② $f^{-1}(1)$ ③ $f^{-1}(-1)$. ④ $f^{-1}(31)$

* في هذا السؤال لانستطيع ايجاد $f^{-1}(x)$ الا من خلال التخمين.

$\boxed{f(a) = b \Leftrightarrow f^{-1}(b) = a}$

solution: By Geussing
بالتخمين

∘∘ solution ① $f(2) = 11 \Leftrightarrow f^{-1}(11) = 2$ ← كأني مسأل نفسي ماهو العدد الذي أعوضه في الاقتران فيكون الناتج = (11)

② $f(0) = 1 \Leftrightarrow f^{-1}(1) = 0$

③ $f(-1) = -1 \Leftrightarrow f^{-1}(-1) = -1$

④ $f(3) = 31 \Rightarrow f^{-1}(31) = 3$

Integrals التكاملات :-

Defn: A function $F(x)$ is said to be an anti-derivate if $F'(x) = f(x)$.

Ex: Find the anti-derivate of following.

① $f(x) = x^4$ ② $f(x) = \cos(x)$.

③ $f(x) = \sec^2(x)$ ④ $f(x) = e^{2x}$

⑤ $f(x) = \ln x$ تؤجل

so/:

① $\frac{x^5}{5}$ ② $F(x) = \sin(x)$.

③ $F(x) = \tan(x)$ ④ $\frac{e^{2x}}{2}$

Academic Team OF Healing

Batch 16